

STATISTICS & MACHINE LEARNING

Complete Training Notes — Cleaned, Corrected, Structured, and Expanded from the Class Transcripts

Purpose: Beginner-friendly study notes for understanding concepts, project work, revision, and interview preparation.

Course Roadmap

- Statistics fundamentals and why statistics matters in data science
- Descriptive vs inferential statistics
- Population, sample, variable, parameter, statistic, and experiment
- Types of data and levels of measurement
- Measures of central tendency and dispersion
- Data distributions, skewness, outliers, and common probability distributions
- Machine learning fundamentals and learning paradigms
- Regression vs classification and classification types
- End-to-end machine learning lifecycle
- Linear regression: best-fit line, residuals, OLS, gradient descent, and learning rate
- Regression evaluation metrics and model interpretation
- Practical corrections to common classroom simplifications

1. What Is Statistics?

Statistics is the discipline of **collecting, organizing, summarizing, analyzing, and interpreting data** so that we can make evidence-based conclusions or decisions. In data science, statistics helps us understand patterns, quantify uncertainty, compare groups, test assumptions, and build reliable predictive models.

Real-life example: A company has three years of sales data. Statistics can help summarize average sales, identify seasonal patterns, measure variability, test whether a promotion changed sales, and estimate future outcomes.

Why Statistics Matters in Data Science

- **Understand the data:** What values are typical? What values are unusual?
- **Find patterns:** Trends, relationships, clusters, and differences between groups.
- **Measure uncertainty:** A sample does not perfectly represent a population.
- **Test claims:** Determine whether observed differences are likely meaningful or random.
- **Support machine learning:** Feature analysis, assumptions, model evaluation, and experimentation all use statistical thinking.

2. Descriptive vs Inferential Statistics

Type	Main Question	What It Does	Example
Descriptive Statistics	What does the observed data show?	Organizes and summarizes the data you already have.	Average salary, median age, sales chart, standard deviation
Inferential Statistics	What can we conclude about a larger population?	Estimate, test, or infer something about a population using sample data.	Estimating voter preference from a survey; test whether a new drug works

Important correction: If 20 out of 100 sampled people prefer a blue car, saying “20% of the sample prefers blue” is descriptive. Using that sample to estimate the percentage in the entire city—with a confidence interval or margin of error—is inferential.

Hypothesis Testing — Basic Idea

Hypothesis testing is an inferential-statistics framework used to evaluate a claim using sample data.

- **Null hypothesis (H0):** Usually represents no effect, no difference, or the status quo.

- **Alternative hypothesis (H1 or Ha):** Represents the effect or difference we are investigating.
- **Example:** H0: the shampoo does not reduce dandruff. H1: the shampoo reduces dandruff.
- We collect appropriate data and use a statistical test to evaluate how compatible the data is with H0.

Do not say: “We prove the null hypothesis.” Standard statistical testing usually leads us to **reject H0** or **fail to reject H0**; it does not prove a hypothesis with absolute certainty.

3. Core Statistical Terminology

Term	Meaning	Example
Population	The complete group of interest.	All customers of a company.
Sample	A subset selected from the population.	1,000 sampled customers.
Variable	A measurable characteristic.	Age, income, product category.
Observation / Data Point	One recorded unit or row.	One customer's record.
Parameter	A numerical characteristic of a population.	True population mean.
Statistic	A numerical characteristic calculated from a sample.	Sample mean.
Experiment	A structured process used to study an outcome.	A/B test of two website designs.

4. Types of Data

Categorical vs Numerical Data

Data Type	Meaning	Examples
Categorical (Qualitative)	Represents groups, labels, or categories.	Gender category, city, brand, yes/no.
Numerical (Quantitative)	Represents measurable or countable quantities.	Age, salary, height, number of purchases.

Discrete vs Continuous Numerical Data

- **Discrete:** Countable values, often integers. Example: number of children, number of orders, defects per item.
- **Continuous:** Can take values across a continuous range. Example: height, weight, time, distance, temperature.

5. Levels of Measurement

Level	Meaning	Order?	True Zero?	Examples
Nominal	Categories with no inherent ranking.	No	N/A	Country, eye color, car brand.
Ordinal	Categories with a meaningful order, but gaps are not necessarily equal.	Yes	No	Poor / Fair / Good / Excellent.
Interval	Numeric scale with equal intervals but no true absolute zero.	Yes	No	Temperature in °C or °F, calendar years.
Ratio	Numeric scale with equal intervals and a meaningful true zero.	Yes	Yes	Height, weight, age, income.

Important correction: Temperature in Celsius or Fahrenheit is **interval**, not ratio, because 0°C does not mean “no temperature.” Kelvin is a ratio scale because it has an absolute zero.

6. Exploratory Data Analysis (EDA)

EDA is the process of examining a dataset before modeling. The objective is not merely to create charts; it is to understand the data-generating situation, data quality, distributions, relationships, anomalies, and modeling risks.

- Understand the business problem and target variable.
- Inspect rows, columns, data types, units, and definitions.
- Check missing values and duplicates.
- Study distributions and unusual values.
- Analyze relationships among variables.
- Detect leakage, imbalance, skewness, and outliers.
- Decide what transformations or feature engineering may be justified.

7. Measures of Central Tendency

Mean

The mean is the arithmetic average. It uses every value and is useful for roughly symmetric numerical data.

$$\text{Mean} = \text{Sum of all observations} / \text{Number of observations}$$

Median

The median is the middle value after sorting the data. For an even number of observations, it is the average of the two middle values. It is more robust than the mean when extreme outliers are present.

Mode

The mode is the most frequently occurring value. It is useful for categorical data and for identifying the most common value in numerical data.

Measure	Best Used When	Main Limitation
Mean	Numerical data without severe outliers; symmetric distributions.	Sensitive to extreme values.
Median	Skewed numerical data or data containing outliers.	Does not use magnitude of every value.
Mode	Most common category/value is important.	May be multiple modes or no unique mode.

8. Measures of Dispersion

Central tendency tells us where the center is. Dispersion tells us how widely the values are spread.

Range

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

The range is simple but depends only on the two extreme values, so it is highly sensitive to outliers.

Variance

Variance measures the average squared distance of observations from the mean. Squaring prevents positive and negative deviations from cancelling each other.

$$\text{Population variance: } \sigma^2 = \sum (x - \mu)^2 / N$$

$$\text{Sample variance: } s^2 = \sum (x - \bar{x})^2 / (n - 1)$$

Standard Deviation

Standard deviation is the square root of variance. It expresses spread in the same units as the original variable.

$$\text{Standard deviation} = \sqrt{\text{Variance}}$$

Important correction: A **small** standard deviation means observations are concentrated close to the mean. A **large** standard deviation means they are more spread out.

9. Distribution, Symmetry, and Skewness

Symmetric / Normal-Like Distribution

A symmetric distribution has similar shape on both sides of its center. In a perfectly symmetric unimodal normal distribution, mean = median = mode.

Right-Skewed (Positive Skew)

The long tail extends to the right. A few unusually large values pull the mean upward. A common relationship is: mode < median < mean.

Left-Skewed (Negative Skew)

The long tail extends to the left. A few unusually small values pull the mean downward. A common relationship is: mean < median < mode.

Key rule: Name the skew by the direction of the **tail**, not by where most observations are located.

10. Outliers

An outlier is an observation that is unusually distant from the main body of the data. Outliers may be errors, rare but valid events, fraud, exceptional cases, or evidence that the population contains multiple subgroups.

- Never remove an outlier only because it looks extreme.
- First verify data quality and business meaning.
- Possible treatments include keeping it, correcting an error, capping/winsorizing, transforming the variable, using robust models, or analyzing the case separately.
- Box plots and the IQR rule are common detection tools.

$$\text{IQR} = Q3 - Q1; \text{ common fences: } Q1 - 1.5 \times \text{IQR} \text{ and } Q3 + 1.5 \times \text{IQR}$$

11. Common Probability Distributions

Distribution	Use Case	Example
Normal (Gaussian)	Continuous measurements clustering around a mean.	Measurement error, approximately normal biological measurements.
Bernoulli	One trial with two possible outcomes.	One coin toss; conversion yes/no.
Binomial	Number of successes in n independent Bernoulli trials with constant probability.	Number of heads in 10 fair coin tosses.
Uniform	Every value/outcome in the defined range has equal probability.	Ideal fair die for discrete uniform outcomes.
Poisson	Count of events occurring in a fixed interval under specific assumptions.	Call center calls arriving per minute.

12. What Is Machine Learning?

Machine learning is a field of AI in which algorithms learn patterns from data and use those patterns to make predictions, classifications, rankings, recommendations, or decisions without requiring a human to manually write a separate rule for every possible case.

Why it became useful: A human may inspect a small dataset manually, but modern systems can contain millions or billions of records. Machine learning automates repeatable pattern-learning at scale. However, it does not eliminate human responsibility: humans still define the problem, validate data, select metrics, evaluate risks, and monitor the system.

13. Main Learning Paradigms

Type	Training Signal	Typical Tasks	Examples
Supervised Learning	Labeled input-output examples.	Regression and classification.	Linear regression, logistic regression, etc.
Unsupervised Learning	No target label is supplied.	Clustering, dimensionality reduction, etc.	K-means, PCA, hierarchical clustering
Semi-Supervised Learning	Small labeled set plus larger unlabeled set	Tasks where labeling is expensive.	Image/text classification with limited labels
Reinforcement Learning	Agent receives rewards from interactions	Sequential decision making.	Game playing, robotics, control systems

Important correction: Unsupervised learning does not normally “predict regression or classification labels” without labels. Its primary goal is to discover structure such as clusters, latent representations, or lower-dimensional patterns.

14. Regression vs Classification

Problem Type	Output	Examples
Regression	Continuous numerical value.	House price, temperature, demand, delivery time.
Classification	Class/category label or class probability.	Spam/not spam, disease class, fraud/not fraud.

Classification Types

- **Binary classification:** Exactly two classes, e.g., fraud vs not fraud.
- **Multiclass classification:** One class selected from more than two mutually exclusive classes, e.g., cat vs dog vs bird.
- **Multilabel classification:** One observation may have multiple labels simultaneously, e.g., an image tagged both “dog” and “outdoor.”

Important correction: Multiple separate images labeled cat, dog, and bird are a **multiclass** dataset if each image has one class. A single image containing both a cat and dog can be **multilabel** if both labels apply.

15. End-to-End Machine Learning Lifecycle

1. **Problem Definition:** Translate the business need into a measurable ML problem. Define the target, prediction unit, constraints, success criteria, and baseline.
2. **Data Collection:** Identify relevant sources and ensure the data is legally, ethically, and technically usable.
3. **Data Understanding:** Understand schemas, definitions, time ranges, target construction, and data quality.
4. **EDA:** Analyze distributions, missingness, relationships, imbalance, outliers, leakage, and temporal effects.
5. **Data Preparation:** Clean data, handle missing values, encode categories, transform variables, and engineer features.

- 6. Train/Validation/Test Strategy:** Split data correctly. For time-dependent problems, use time-aware validation rather than random splitting.
- 7. Model Selection:** Choose algorithms appropriate for the task, data size, interpretability needs, latency, and constraints.
- 8. Training:** Fit the model on training data.
- 9. Evaluation:** Evaluate on validation/test data using metrics aligned with business cost.
- 10. Hyperparameter Tuning:** Tune model settings without contaminating the final test set.
- 11. Packaging:** Save the model and all required preprocessing artifacts. Pickle is one option, not the only production format.
- 12. Deployment:** Serve predictions through batch jobs, streaming pipelines, or APIs.
- 13. Monitoring:** Track performance, drift, latency, errors, fairness, and data quality.
- 14. Retraining:** Retrain when justified by drift, new data, business changes, or scheduled governance.

16. Linear Regression

Linear regression models the relationship between one or more input variables and a continuous target. In simple linear regression, the model fits a straight line.

$$\hat{y} = mx + c$$

- $\hat{y}$ (**y-hat**): predicted target value.
- x : input feature.
- m : slope/coefficient — how much the prediction changes for a one-unit change in x .
- c : intercept — predicted y when $x = 0$, if that value is meaningful in the domain.

Actual Value, Predicted Value, and Residual

A residual is the difference between an observed value and the value predicted by the model.

$$\text{Residual: } e_i = y_i - \hat{y}_i$$

The goal is not literally to make every residual zero—this is usually impossible and may indicate overfitting. The goal is to estimate a relationship that generalizes well to unseen data.

17. Ordinary Least Squares (OLS)

OLS estimates linear-regression coefficients by minimizing the **sum of squared residuals**.

$$\text{SSE} = \sum (y_i - \hat{y}_i)^2$$

For the matrix formulation, when the required inverse exists, the closed-form solution can be written as:

$$\beta = (X^T X)^{-1} X^T y$$

Important correction: The closed-form normal equation is not inherently an iterative process. Gradient-based optimization is iterative; the standard closed-form OLS solution is computed directly through linear algebra. In practice, robust numerical methods such as QR or SVD decompositions are often preferred over explicitly calculating a matrix inverse.

18. Gradient Descent

Gradient descent is an iterative optimization algorithm. It starts with parameter values, calculates the gradient of the loss function, and updates the parameters in the direction that reduces loss.

$$\theta := \theta - \alpha \nabla J(\theta)$$

- θ : model parameters.
- α : learning rate.
- $J(\theta)$: loss/cost function.
- $\nabla J(\theta)$: gradient indicating the direction of steepest increase; subtracting it moves downhill.

Learning Rate

- **Too small**: training can be very slow.
- **Too large**: optimization can overshoot, oscillate, or diverge.
- **Appropriate**: reaches a good minimum efficiently.

Nuance: For ordinary linear regression with mean squared error, the loss surface is convex, so there is a single global minimum rather than problematic local minima. Local-minimum discussions become more relevant in non-convex models such as deep neural networks.

19. Regression Evaluation Metrics

Mean Absolute Error (MAE)

$$\text{MAE} = (1/n) \sum |y_{\text{actual}} - y_{\text{predicted}}|$$

MAE is the average absolute prediction error. It is easy to interpret because it is expressed in the target's original units. Compared with squared-error metrics, it is **less sensitive** to large outliers.

Mean Squared Error (MSE)

$$\text{MSE} = (1/n) \sum (y_{\text{actual}} - y_{\text{predicted}})^2$$

MSE squares errors, so large errors receive a much larger penalty.

Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\text{MSE}}$$

RMSE keeps the stronger penalty for large errors while returning the result to the original target units.

R-Squared (R^2)

$$R^2 = 1 - [\sum (y_{\text{actual}} - y_{\text{predicted}})^2 / \sum (y_{\text{actual}} - \bar{y})^2]$$

R^2 measures how much of the target variance is explained by the model relative to a baseline that predicts the mean. An R^2 of 0.75 is commonly interpreted as the model explaining about 75% of the observed variance in the target.

Important: A high R^2 does not prove that the model is correct, causal, unbiased, or useful on new data. R^2 can also be negative on evaluation data when the model performs worse than predicting the mean.

20. MAE vs MSE vs RMSE vs R^2

Metric	Interpretation	Sensitivity to Large Errors	Units
MAE	Average absolute error.	Lower	Same as target
MSE	Average squared error.	High	Squared target units
RMSE	Square root of average squared error.	High	Same as target
R^2	Relative variance explained vs mean baseline	Indirect	Unitless

21. AI, ML, Deep Learning, NLP, and Generative AI

These areas overlap, but they should not be treated as completely isolated boxes.

Area	Main Focus
Artificial Intelligence (AI)	Broad field of building systems that perform tasks associated with intelligent behavior.
Machine Learning (ML)	Learning patterns from data for prediction or decision-making.
Deep Learning (DL)	Machine learning based on multi-layer neural networks; used for structured and unstructured data.
Natural Language Processing (NLP)	Methods for processing, understanding, and generating human language.
Generative AI	Models that generate new content such as text, images, audio, video, or code.
Agentic AI	Systems that use models, tools, memory, planning, and actions to pursue goals across multiple steps.

Important correction: Machine learning is not limited to structured/tabular data, and deep learning is not limited to images. ML and deep learning can work with text, images, audio, time series, graphs, and tabular data. Modern generative AI often uses deep learning and NLP together.

22. Practical Project Checklist

- Write a clear problem statement before selecting an algorithm.
- Define the target and prediction unit precisely.
- Understand every important feature and its unit.
- Separate data cleaning from modeling logic.
- Check missing values, duplicates, outliers, skewness, and leakage.
- Use appropriate visualizations to support—not replace—analysis.
- Choose train/validation/test splitting based on how the model will be used in real life.
- Start with a simple baseline before complex models.
- Use metrics aligned with business consequences.
- Save preprocessing together with the model.
- Document assumptions and limitations.
- Monitor production models; deployment is not the end of the ML lifecycle.

23. Quick Interview Revision

Q: What is descriptive statistics?

It summarizes and describes the data we already observed using measures such as mean, median, standard deviation, tables, and charts.

Q: What is inferential statistics?

It uses sample data to make estimates or test claims about a larger population while accounting for uncertainty.

Q: When should you prefer median over mean?

When the numerical distribution is strongly skewed or contains influential outliers.

Q: What is standard deviation?

It measures the typical spread of observations around the mean and is expressed in the original variable's units.

Q: What is supervised learning?

Learning a mapping from inputs to known target labels using labeled training examples.

Q: Regression vs classification?

Regression predicts continuous values; classification predicts categories or class probabilities.

Q: What is a residual?

The difference between the actual target value and the model's predicted value.

Q: What does OLS do?

It estimates linear-regression coefficients by minimizing the sum of squared residuals.

Q: What is gradient descent?

An iterative optimization method that updates model parameters in the direction that reduces a loss function.

Q: MAE vs RMSE?

MAE gives average absolute error and is more robust to large errors; RMSE penalizes large errors more strongly.

24. Final Concept Map

Statistics helps us understand data and uncertainty → **EDA** reveals quality, distributions, and relationships →

Machine Learning learns patterns for prediction or decision-making → **Evaluation** tells us whether the model generalizes → **Deployment and Monitoring** turn the model into a maintained real-world system.

The most important habit is not memorizing algorithms. It is learning to ask: **What is the problem? What does the data actually represent? What assumptions am I making? What metric reflects real success? Could the model fail when the real-world distribution changes?**

Source basis: Prepared from the two uploaded class transcript files. Repetition and speech fillers were removed, concepts were reorganized, and several technical inaccuracies were corrected for study use.